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① A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. Explain the different types of intelligent agents and justify the best approach.

⇒ A smart Home Automation System uses intelligent agents to control lights, temperature, and security by analyzing inputs such as motion detection, temperature, and time of day.

1. Simple Reflex Agent :-

- * Acts only on the current input using if-then rules.

- * Does not store previous information.

Eg :-

If motion detected at night → Turn on the lights.

2. Model-Based Agent :-

- * Maintains an internal memory of previous states.

- * Uses both current and past information.

Eg :- It switches off light automatically.

3. Goal-Based Agent :-

- * Takes actions to achieve a specific goal.
- * And its example is maintain room temperature between 22°C - 24°C and lock all doors at night.

4. Utility-Based Agent :-

- * Chooses the action that provides the maximum overall benefit and considers comfort, energy saving, and security together.
- * And its example is Adjusts the AC temperature to reduce electricity usage while keeping the room comfortable.

There is a best Approach (Justification)

A Hybrid Agent (Model-Based + Goal-Based + Utility-Based) is the best choice because :-

- * Model-Based agent remembers user preferences.
- * Goal-Based Agent achieves comfort and safety goals.
- * Utility-Based agent balances comfort, energy efficiency, and security.

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Conclusion :-

A hybrid intelligent agent is the most suitable for a smart home automation system because it learns user behavior, achieves predefined goals, and optimizes comfort, energy efficiency, and security. This makes the system more intelligent, reliable, and user-friendly.

② A robot is placed in an unknown maze and must find a path to exit without prior knowledge of the layout. Explain how Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) solve this problem. Discuss their advantages, disadvantages, and relate them to intelligent agents.

⇒ A robot placed in an unknown maze must explore different paths to reach the exit. Since it has no prior knowledge of the maze, it uses unfocused search algorithms such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

4. Uniform Cost Search (UCS) :-

- * UCS expands the path with the lowest cumulative cost first.
- * It guarantees that least-cost path to the exit.

Eg :-

If different paths have different travel costs, UCS selects the cheapest path.

2. Iterative Deepening Search (IDS):

- * IDS combines the advantages of Depth-First Search (DFS) and Breadth-First Search (BFS).
- * It repeatedly searches one level with ~~inc~~ increasing depth limits until the goal is found.

Eg :-

The robot first searches one level, then two levels, and so on until it finds the exit.

Advantages :-

- * Finds the optimal solution and works well when path costs are different.
- * Uses very little memory and finds the shortest path when all step costs are equal.

Disadvantages :-

- * Requires high memory and slower for large search spaces.
- * Repeats searches at each depth, and can take more time due to repeated exploration.

Relation to Intelligent Agents :-

- * ~~Simple Reflex Agent~~ : Acts only on the ~~current~~ situation and may get stuck in the maze.
- * ~~Model-Based Agent~~ : Remembers visited paths and avoids revisiting the same locations.
- * ~~Goal-Based Agent~~ : Searches for actions that lead to the exit.
- * ~~Utility-Based Agent~~ : Chooses the best path by considering factors such as distance, time, and energy.

Best combination :-

A Goal-Based Agent with Uniform Cost Search (UCS) is the most effective combination because :-

- * The Goal-Based Agent focuses on reaching the exit.
- * UCS guarantees the lowest-cost path, making the robot efficient and reliable.

If memory is limited, a Goal-Based Agent with Iterative Deepening Search (IDS) is a good alternative because IDS uses less memory.

Conclusion :-

For solving an unknown maze, UCS is best when finding the optimal path is important while IDS is suitable when memory is limited. A Goal-Based Agent combined with UCS provides the most effective solution because it intelligently plans actions and finds the optimal path to the exit.